



Descriptives Statistics & Probability Distributions

Lecture 2 - LAS

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Today

- Descriptives Statistics
- Probability
- Discrete probability distributions
- Continuous probability distributions
- Describing data



Today

[Chapter 2](#) and [Chapter 4](#) from the gitbook.

❗ Mistake in slide from last week!

- Deadline assignment #1 is Sunday October 18th.
- Deadline on Canvas is always guiding!



Descriptives Statistics

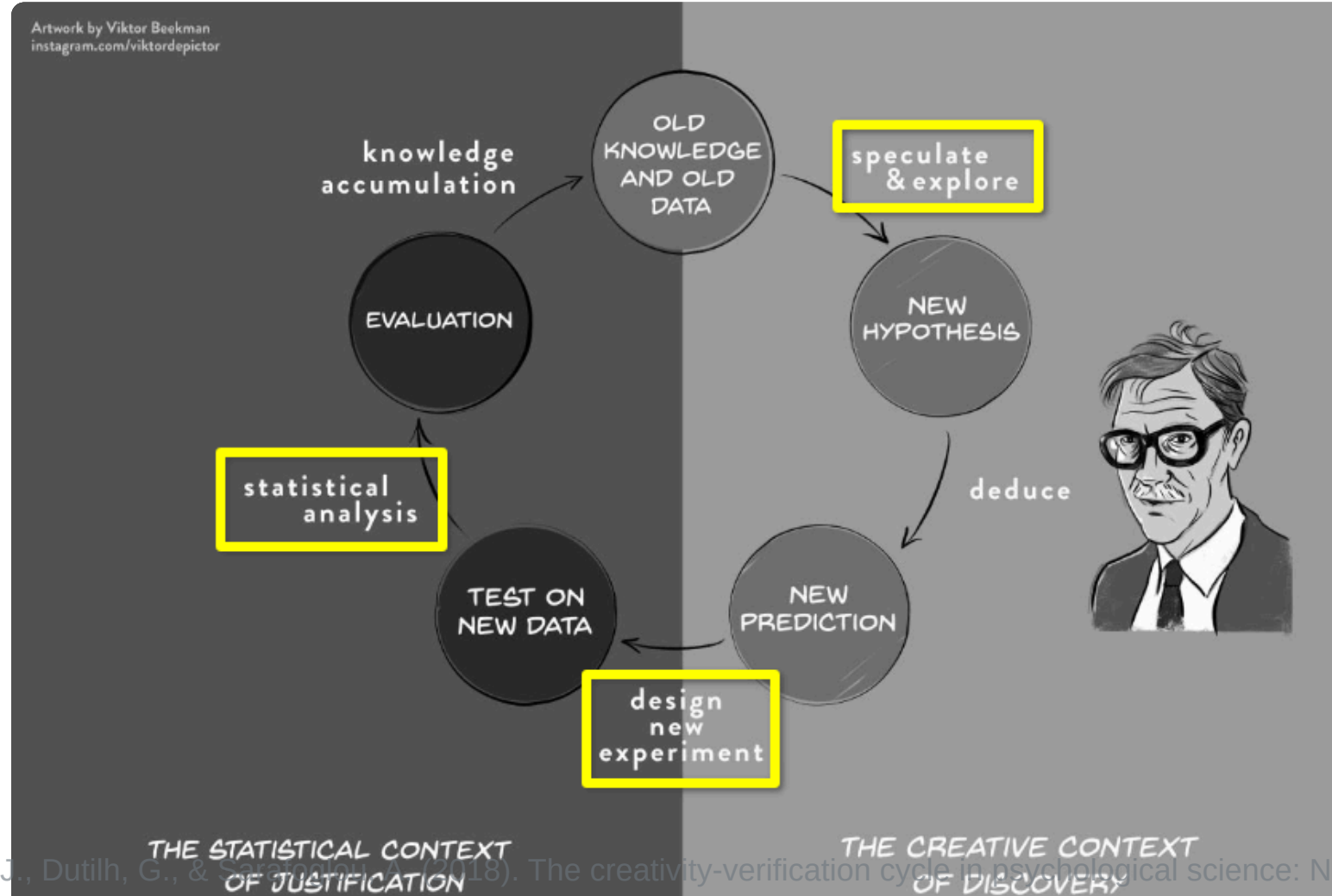


Why do We use Descriptives Statistics?

- to describe the sample
- to check for mistakes (attention check)
- to check assumptions
- to answer research questions that do not require hypothesis tests



When do We use Descriptives Statistics?



Source: Wagenmakers, E. J., Dutilh, G., & Sarason, D. (2018). The creativity-verification cycle in psychological science: New methods to combat old idols. *Psychological Science*, 29(4), 418-427. <https://doi.org/10.1177/1745691618771357>



Descriptives Statistics Today

- Central Tendency
- Dispersion/ Spread/ Variability
- Skewness
- Kurtosis

Data Example: Bechdel Test





Data Example: Bechdel Test

YEAR	IMDB	TITLE	TEST	CLEAN_TEST	BINARY	BUDGET	DC
2013	tt1711425	21 & Over	notalk	notalk	FAIL	13000000	25
2012	tt1343727	Dredd 3D	ok-disagree	ok	PASS	45000000	13
2013	tt2024544	12 Years a Slave	notalk-disagree	notalk	FAIL	20000000	53
2013	tt1272878	2 Guns	notalk	notalk	FAIL	61000000	75
2013	tt0453562	42	men	men	FAIL	40000000	95



Measures of Central Tendency

The most “common” value, one “representative” number

- average student
- most popular show on Netflix
- proportion of movies that passes the bechdel test
- average movie budget
- most common bechdel test result



Central Tendency: Mean

$$\bar{x} = \frac{\sum_{i=1}^n x_i}{n} = \frac{x_1 + x_2 + \dots + x_n}{n}$$

Definitions:

- x : the variable we're interested in.
- $\bar{x}$: the sample mean of the variable
- x_i : the i th observation of the variable.
- n the sample size.
- $\sum_{i=1}^n$: the sum from i is 1 to i is n .



Central Tendency: Mean

Let's call "PASS" 1 and "FAIL" 0.

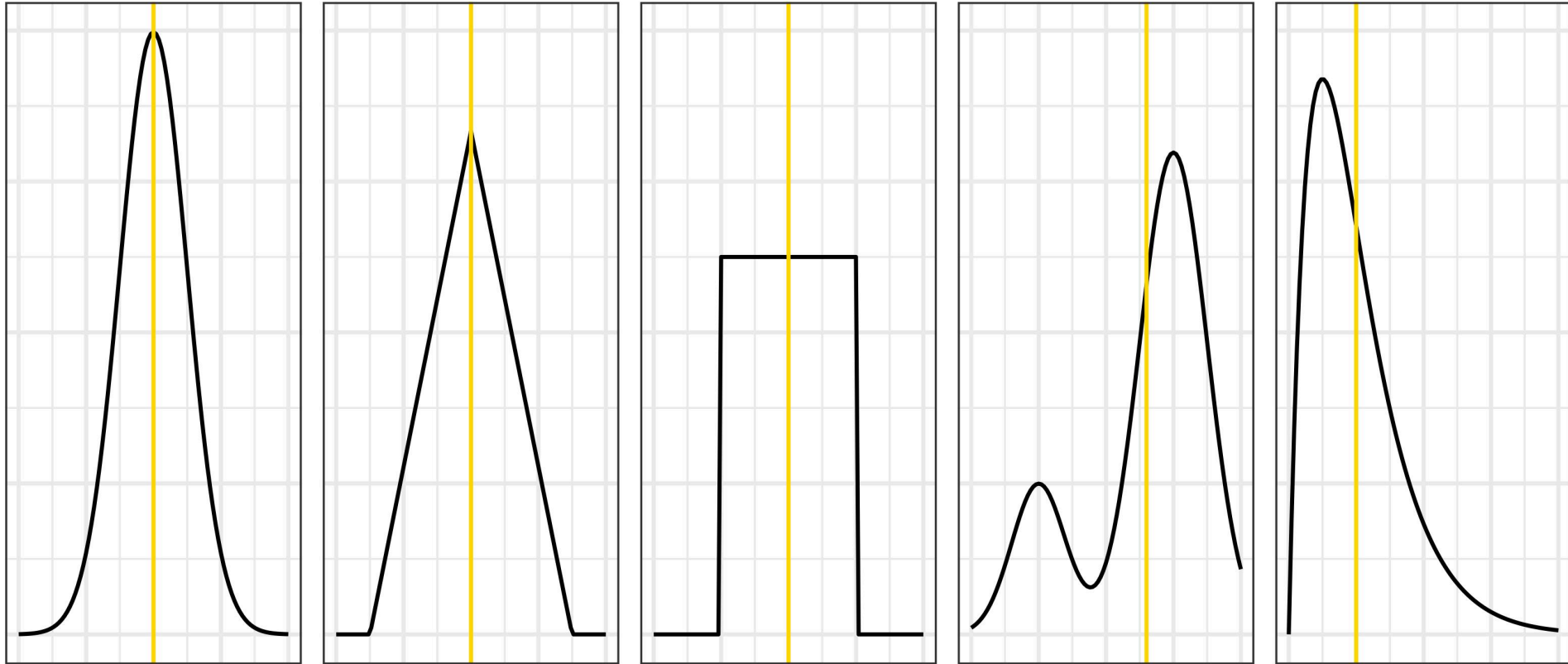
What does the mean of these numbers compute?

The proportion of movies that passed the Bechdel test.

No. "PASS" (803) / No. Movies (1794) = 0.448



Central Tendency: Mean





Central Tendency: Median

Definition: Median

If you were to order all scores of your variable from lowest to highest, then the median value is the value that splits your sample in half: half of the participants score lower than this value, and half score higher.

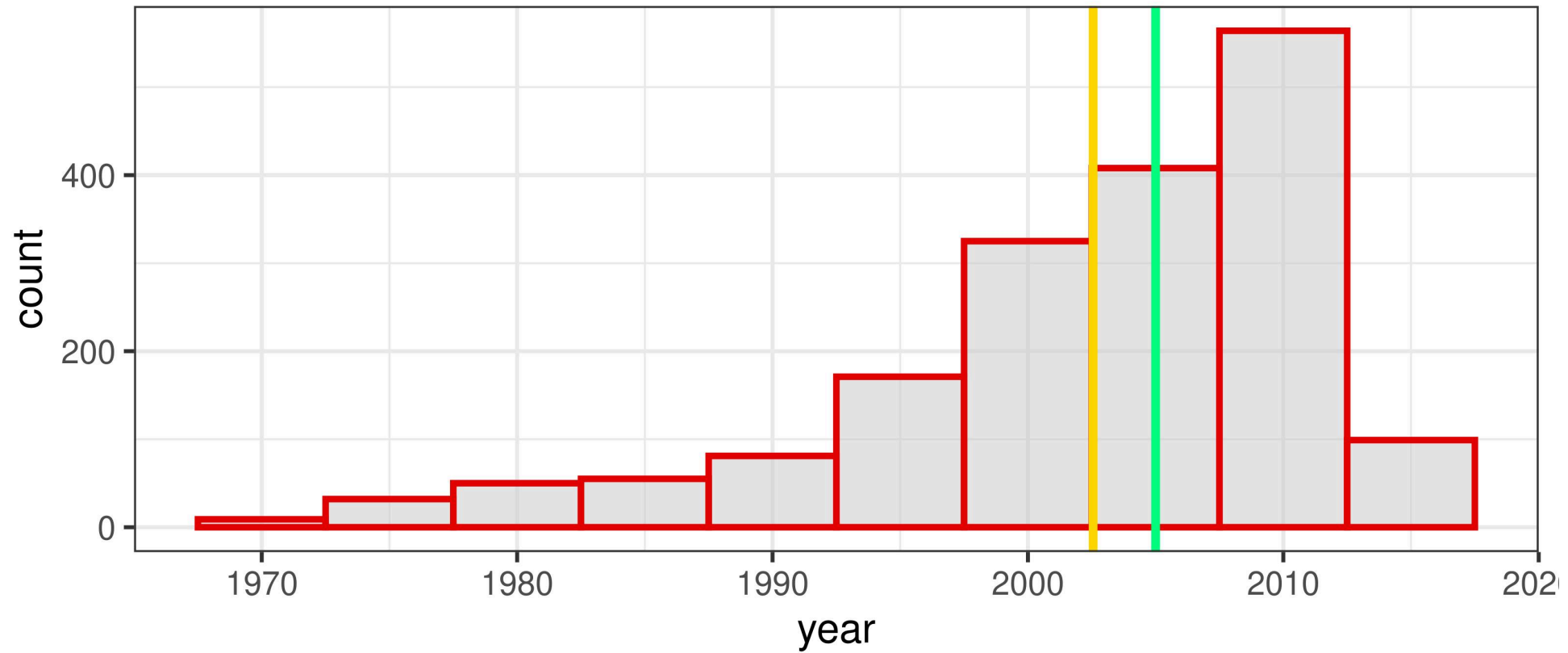
- Another name for the median is the 50th percentile. The nth percentile is the score that divides the sample so that % of participants score lower.
- If the distribution is symmetric, then the median equals the mean

Central Tendency: Median

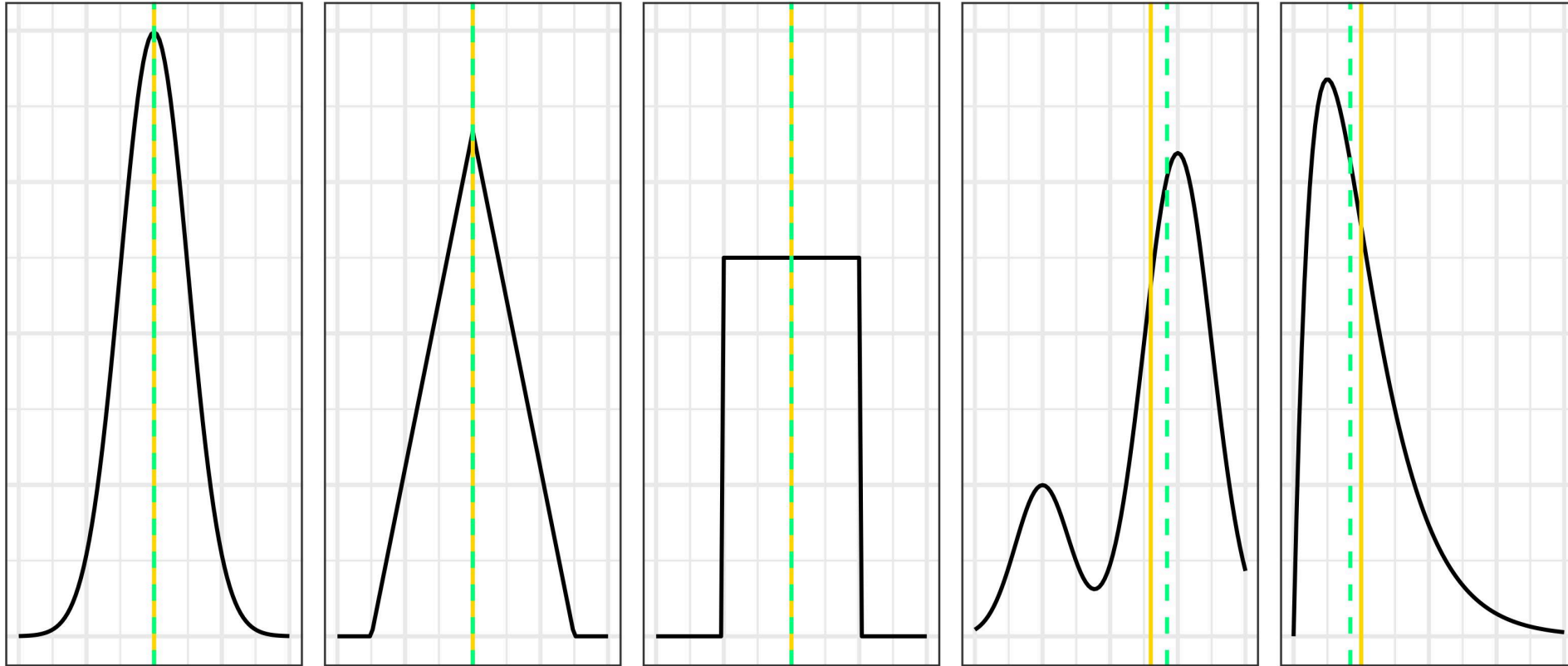


Year	Freq
1970	1
1971	5
1972	3
1973	5
1974	7
1975	5
1976	8
1977	7
1978	8
1979	5
1980	14
1981	9
1982	14

Central Tendency: Median



Central Tendency: Median



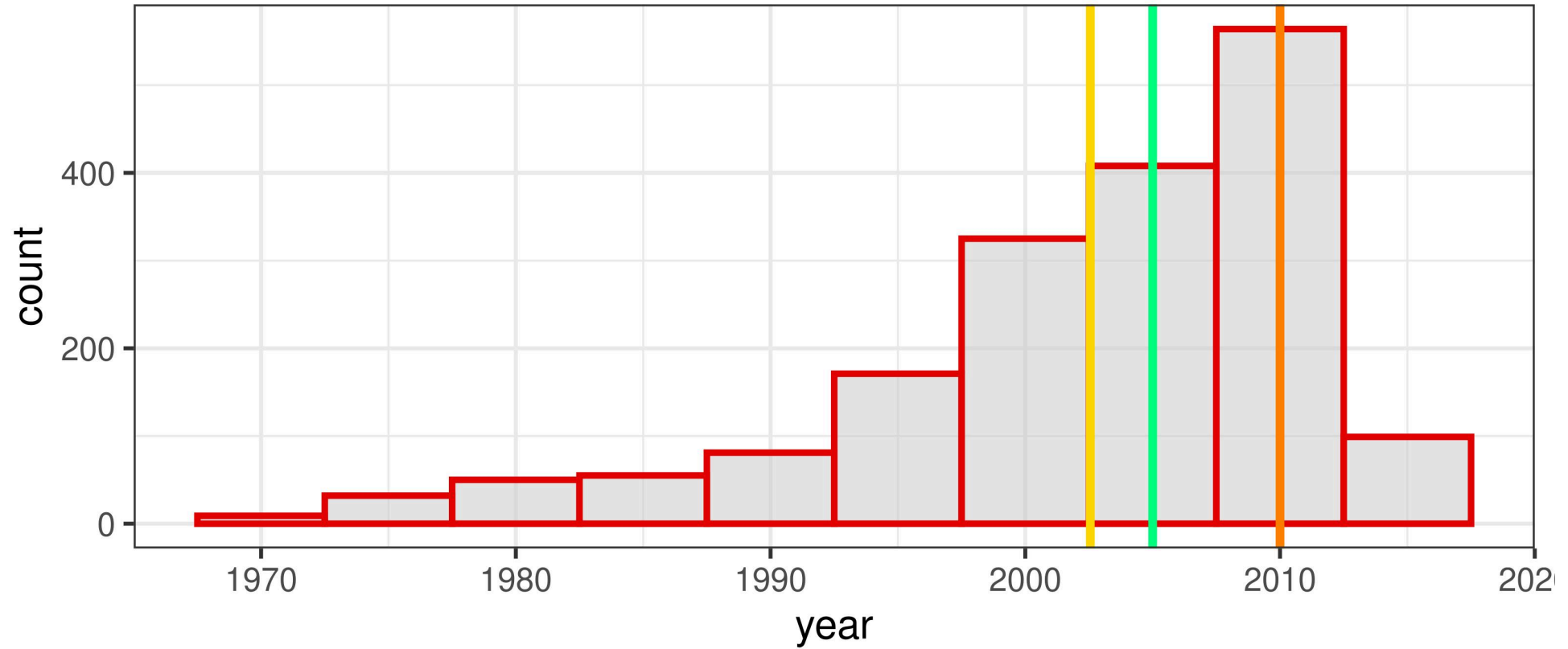
Central Tendency: Mode



- The most common value



Central Tendency: Mode





Dispersion

- Range
- Sum of Squared Distances to the Mean
- Variance: mean squared distance
- Standard Deviation
- Mean absolute deviation (MAD)



Range

$$R = x_{\text{largest}} - x_{\text{smallest}}$$

$$R_{\text{year}}: 2013 - 1970 = 43.$$



Sum of Squared Distances to the Mean: SS_x

$$\bar{x} = \frac{\sum_{i=1}^n x_i}{n} = \frac{x_1 + x_2 + \dots + x_n}{n}$$

$$SS_x = \sum_{i=1}^n (x_i - \bar{x})^2 = (x_1 - \bar{x})^2 + (x_2 - \bar{x})^2 + \dots + (x_n - \bar{x})^2$$

- $\overline{\text{year}} = 2002.5523969$
- $SS_{\text{year}} = 144579.6$



Variance: mean squared distance

$$\bar{x} = \frac{\sum_{i=1}^n x_i}{n} = \frac{x_1 + x_2 + \dots + x_n}{n}$$

$$SS_x = \sum_{i=1}^n (x_i - \bar{x})^2 = (x_1 - \bar{x})^2 + (x_2 - \bar{x})^2 + \dots + (x_n - \bar{x})^2$$

$$s_x^2 = \frac{SS_x}{n}$$

- $\overline{\text{year}} = 2002.5523969$
- $SS_{\text{year}} = 144579.6$
- $s_{\text{year}}^2 = 80.5906213$



Standard Deviation: root mean squared distance

$$\bar{x} = \frac{\sum_{i=1}^n x_i}{n} = \frac{x_1 + x_2 + \dots + x_n}{n}$$

$$SS_x = \sum_{i=1}^n (x_i - \bar{x})^2 = (x_1 - \bar{x})^2 + (x_2 - \bar{x})^2 + \dots + (x_n - \bar{x})^2$$

$$s_x^2 = \frac{SS_x}{n}$$

$$s_x = \sqrt{s_x^2}$$

- $\overline{\text{year}} = 2002.5523969$
- $SS_{\text{year}} = 144579.6$
- $s_{\text{year}}^2 = 80.5906213$
- $s_{\text{year}} = 8.9772279$



Mean absolute deviation (MAD)

$$\text{MAD}_x = \frac{\sum_{i=1}^n |x_i - \bar{x}|}{n} = \frac{|x_1 - \bar{x}| + |x_2 - \bar{x}| + \dots + |x_n - \bar{x}|}{n}$$

- $\text{MAD}_{\text{year}} = 6.9458004$



Skewness and Kurtosis

- Skewness governs symmetry
 - positive skew / right tailed / larger observations *more* likely than smaller observations
 - negative skew left tailed / larger observations *less* likely than smaller observations
 - no skew / symmetric / / larger observations *equally* likely as smaller observations
- Kurtosis governs extreme values
 - High kurtosis: extreme values are rare relative to values near the median
 - Low kurtosis: extreme values are common relative to values near the median



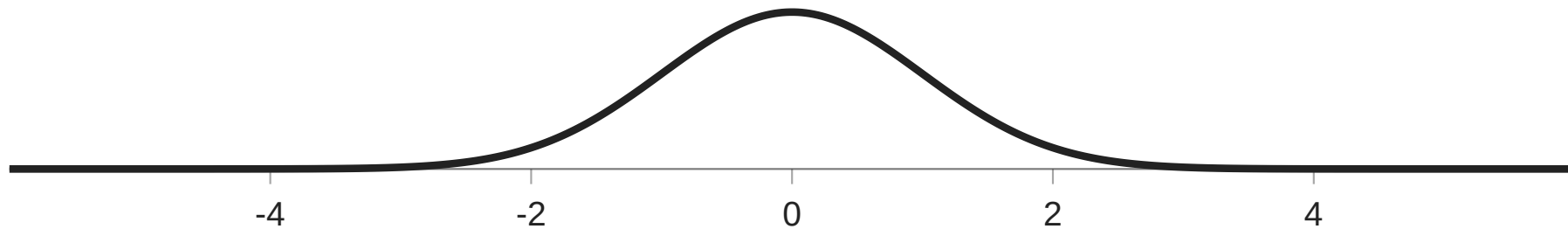
Skewness and Kurtosis

Mean 0.0

SD 1.0

Skewness 0.0

Kurtosis 1.0





What have we learned?

- Most movies did not pass the Bechdel test.
- Most movies are from 2000 - 2010.
- The distribution of years is *left skewed*.



Probability



Population versus Sample

- The “Descriptive Statistics” before were doing inference on a sample.
- We will now see that each of these has a population counterpart.
- For this, we first need to some probability theory.



Defining probability

Probability: How likely something is to occur.

- Specifically, the proportion of times we expect to observe an outcome in a *random experiment* that could be repeated many times

Random experiment

- Could theoretically be repeated an infinite number of times
- Before I flip a coin, the coinflip is a random experiment with $P_{heads} = .5$, or 50%
- After I flip the coin, it's no longer random; it's fixed (either 100% or 0%)



Examples probability

Probability:

- of a baby being born male, $p_{male} = .51$
- of rolling a 7 with two dice in Catan, $p_7 = .167$

Not probability:

- The probability that there is life on other planets is either 100% or 0%
- Even if you don't know which is correct



Random variables

Random variable: An unknown value that follows some probability distribution.

Disambiguation: Last lecture we used “variable” as a placeholder for some unknown number.

- The outcome of a coin toss is a random variable
- It is discrete because it has only two possible outcomes
- The probability distributions is
 - $P(X = \text{heads}) = 0.5$
 - $P(X = \text{tails}) = 0.5$
 - Together: 1.0, so all possible outcomes are covered

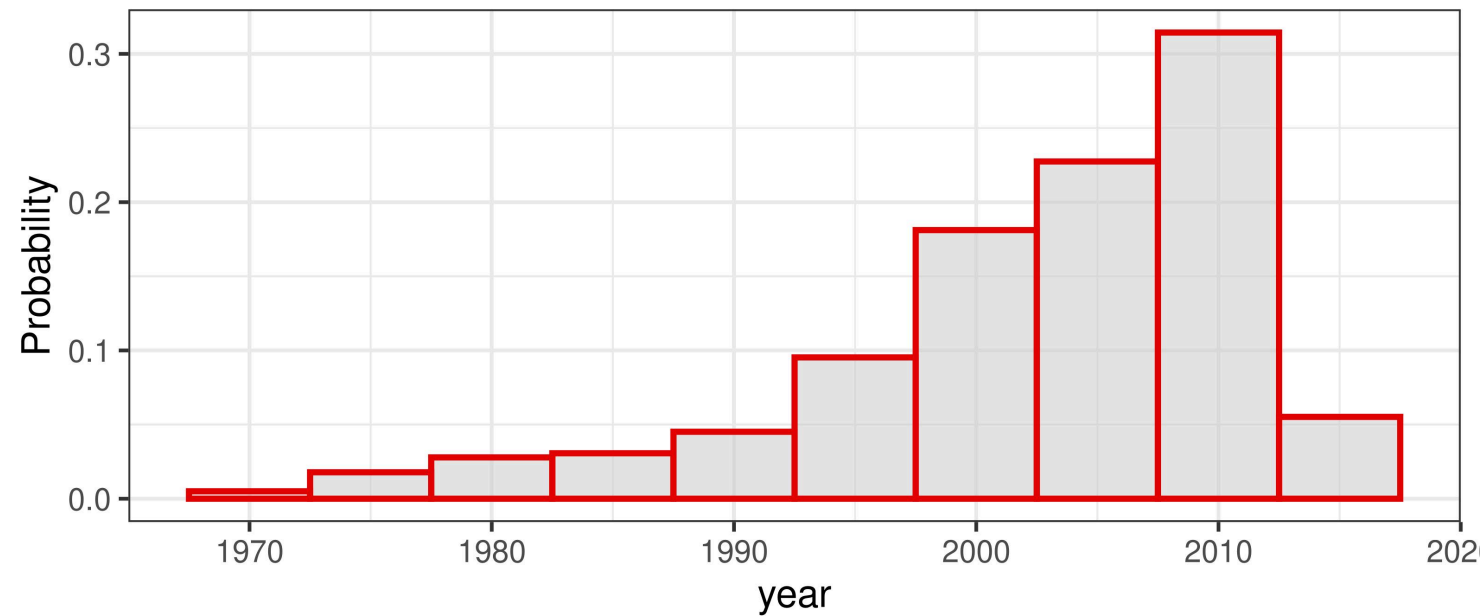


Break

See you in ~10 minutes!



Discrete probability distributions





Frequency vs probability

Frequency distributions

- Summarize observed outcomes in a sample
- E.g., the number of Dutch/foreign students in this class

A probability distribution is similar, but

- Can be interpreted as the (estimated) probability of observing these outcomes in the future
- E.g., if I select a random student in this class, what's the probability that they will be Dutch?



Discrete distributions

- Probability mass function (pmf)
 - 100% of the probability mass is distributed among finite discrete outcomes
- Discrete => Finite number of outcomes
 - Dutch student / foreign student
 - Student has tattoos / no tattoos
 - Movie passes/ fails the Bechdel test
 - Describe using contingency table or bar chart

	Gender			Total
	Female	Male	Mixed/other	
Test				
FAIL	21	957	13	991



Indexing contingency tables

i : row

j : column

f : frequency (number of observations)

$f_{i,j}$: frequency in the cell in row i , column j

\$J\$			
i	$j = 1$	$j = 2$	MARGIN ROWS
$i = 1$	$f_{1,1}$	$f_{1,2}$	$f_{1,+}$
$i = 2$	$f_{2,1}$	$f_{2,2}$	$f_{2,+}$
Margin columns	$f_{+,1}$	$f_{+,2}$	$f_{+,+}$



Marginal frequency distribution

Separate frequency distribution of each variable in contingency table

i	j		MARGIN ROWS
	$j = 1$	$j = 2$	
$i = 1$	$f_{1,1}$	$f_{1,2}$	$f_{1,+}$
$i = 2$	$f_{2,1}$	$f_{2,2}$	$f_{2,+}$
Margin columns	$f_{+,1}$	$f_{+,2}$	$f_{+,+}$



Marginal frequency distribution

For the variable “Gender”: How female directors?

For the variable “Test”: How many movies passed?

TEST	DIRECTOR GENDER			TOTAL ROW
	FEMALE	MALE	MIXED/OTHER	
FAIL	21	957	13	991
PASS	94	689	20	803
Total column	115	1646	33	1794



Conditional frequency distribution

Frequency distribution of one variable, for specific value of the other variable

E.g.: What is the conditional frequency distribution of Gender **for movies that passed?**

TEST	DIRECTOR GENDER			TOTAL ROW
	FEMALE	MALE	MIXED/OTHER	
FAIL	21	957	13	991
PASS	94	689	20	803
Total column	115	1646	33	1794



Joint frequency distribution

Frequency of a combination of two (or more) variables

E.g.: What is the frequency of **movies that passed with a male director**?

TEST	DIRECTOR GENDER			TOTAL ROW
	FEMALE	MALE	MIXED/OTHER	
FAIL	21	957	13	991
PASS	94	689	20	803
Total column	115	1646	33	1794



Frequencies to probabilities

Divide frequencies by a total to get a probability distribution

Which frequencies and which total you use depends on what probability distribution you want



Marginal probability distribution

Divide marginal totals by the **global total**

E.g.: What is the **marginal probability distribution** of a movie passing?
 $P(Pass)$

TEST	DIRECTOR GENDER			TOTAL ROW
	FEMALE	MALE	MIXED/OTHER	
FAIL	21	957	13	$991 / 1794 = 0.55$
PASS	94	689	20	$803 / 1794 = 0.45$
Total column	115	1646	33	1794



Conditional probability distribution

Divide the row- or column frequencies by the **marginal total**

E.g.: What is the conditional probability of **passing the test for a movie with a male director**? $P(Pass|Male)$

TEST	DIRECTOR GENDER			TOTAL ROW
	FEMALE	MALE	MIXED/OTHER	
FAIL	21	957	13	991
PASS	94 / 803 = 0.12	689 / 803 = 0.86	20 / 803 = 0.02	803
Total column	115	1646	33	1794



Joint probability distribution

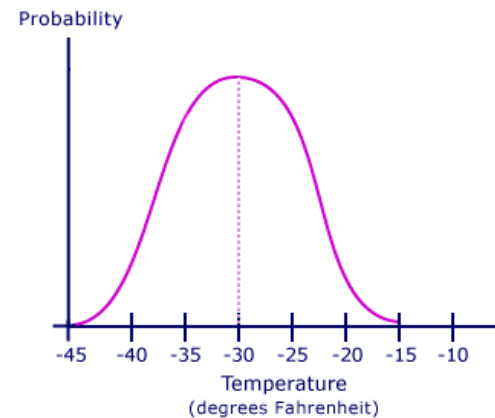
Divide the cell frequency by the **global total**

E.g.: What is the joint probability of a movie **Passing** and having a female **director**? $P(Pass \cap Female)$

TEST	DIRECTOR GENDER			TOTAL ROW
	FEMALE	MALE	MIXED/OTHER	
FAIL	21	957	13	991
PASS	94 / 1794 = 0.05	689	20	803
Total column	115	1646	33	1794



Continuous probability distributions





Continuous probability distributions

- Infinite possible outcomes
- No exact probability for specific values
- Continuous *probability density function* describes how likely each outcome is
 - Cf. probability mass function for discrete outcomes
- Surface area determines probability
- Many probability distributions exist
- This course covers only one...



Normal Distribution

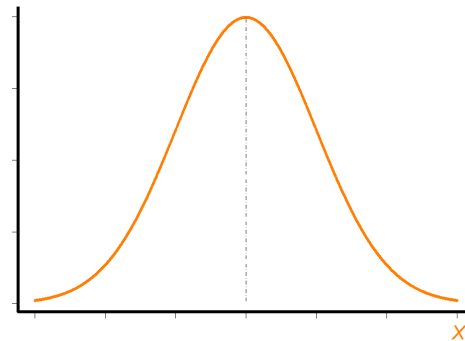
Theoretical distribution for continuous variables

Bell-shaped, symmetric, from -infinity to +infinity

You can use it to describe (=model) real data

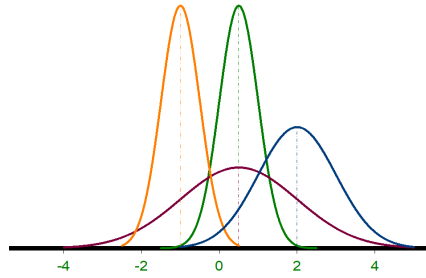
Two parameters:

- Mean μ (“mu”): Most common / average value
- Standard deviation σ (“sigma”): Average deviation from the mean





Examples of Normal Distributions



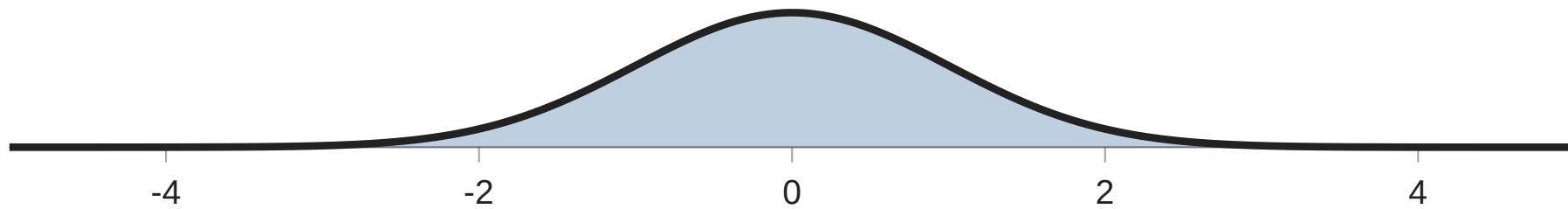
DIST.	μ	σ
1	-1.0	0.5
2	0.5	0.5
3	0.5	1.5
4	2.0	1.0



Normal Demo

μ  0.0

σ  1.0





Standard Normal Distribution (Z-distribution)

$Z \sim N(\mu = 0, \sigma^2 = 1)$ Note $\sigma^2 = \sigma = 1$

- The normal distribution is *closed* under linear transformations.
 - i.e., if $Y \sim N(a, b)$ then $X = c \times Y + d \sim N(c \times a + d, c \times b)$.

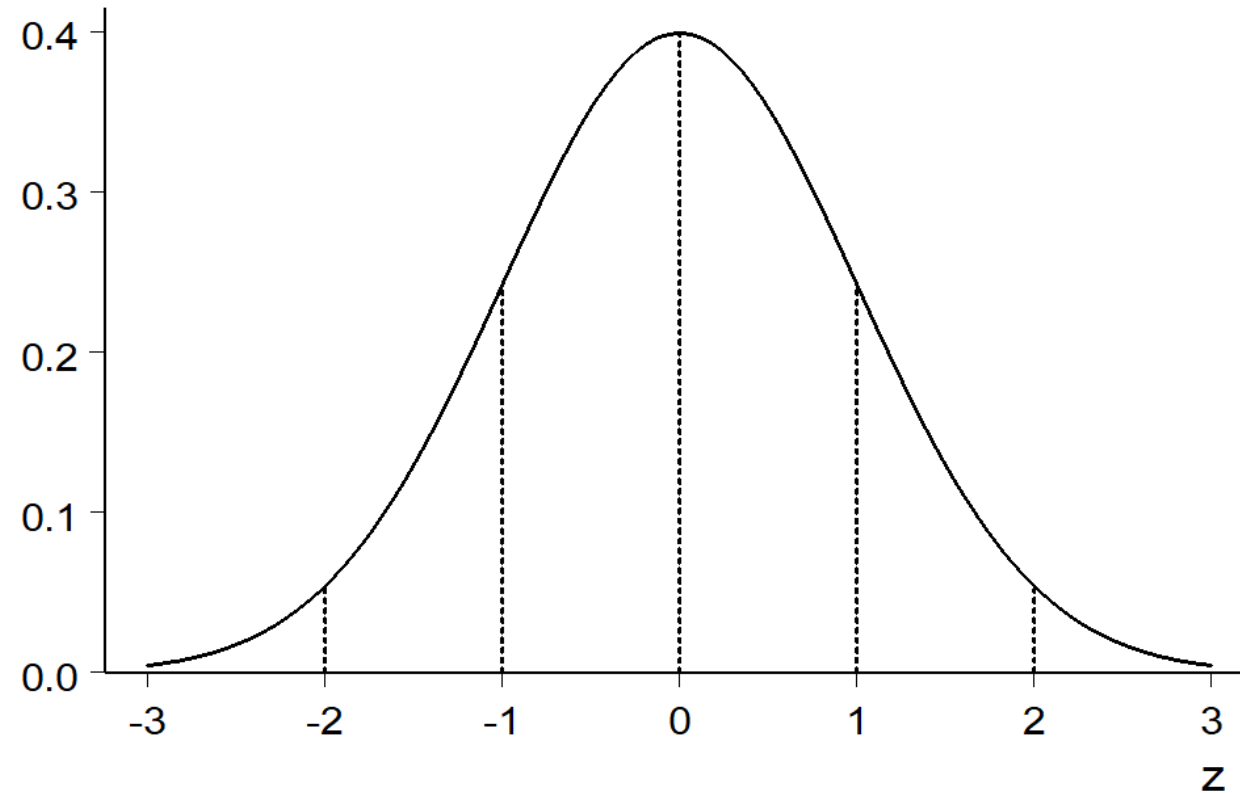
Because of this:

- We can standardize any normal distribution to the Z-distribution
- This removes the units of measurement of our original variable
- Standardizing allows us to calculate probabilities more easily
- Stats books contain probability values for the Z-distribution
- We can always convert back to the original units



Standard Normal Distribution

$$Z \sim N(\mu = 0, \sigma = 1)$$





X to Z and vice versa

You can standardize values of any normally distributed variable

$$Z = \frac{X - \mu_x}{\sigma_x}$$

Standardizing: Removing the original units of measurement

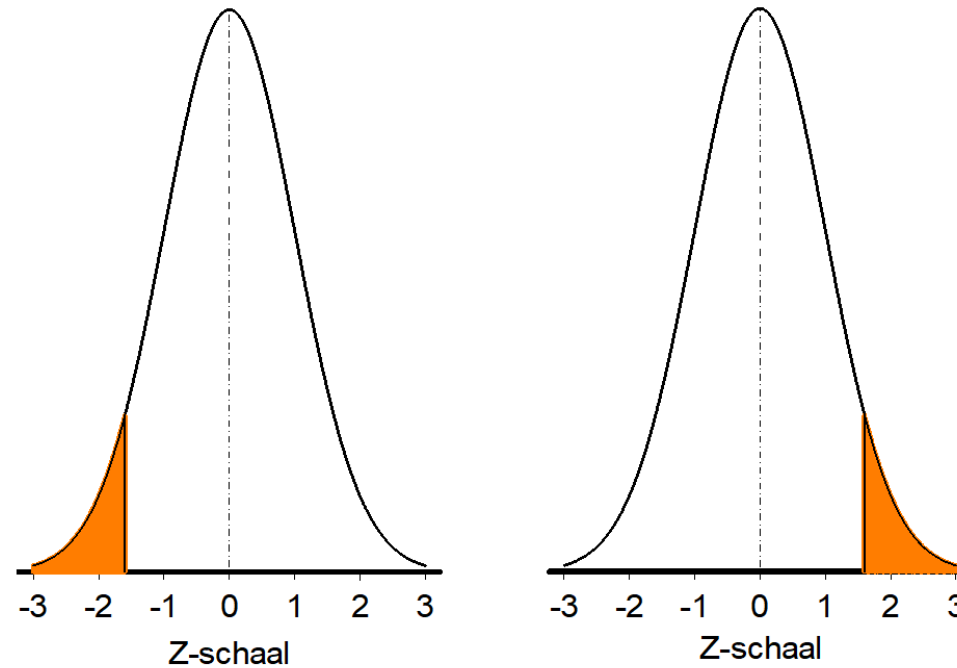
And reverse it to get the units back; for any Z-value:

$$X = \mu_x + (Z \times \sigma_x)$$



Properties of the Normal Distribution (1 / 4)

- The distribution is symmetric



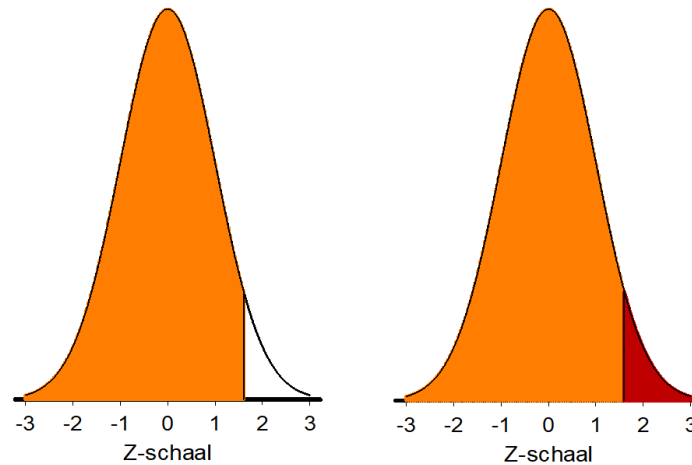
$$P(Z < -1.64) = P(Z > 1.64) = 0.05$$



Properties of the Normal Distribution (2 / 4)

- Total surface area is 1
- We can find areas by taking the complement (1-something)

$$P(Z < 1.64) = .95$$



$$\text{So } P(Z > 1.64) = 1 - .95 = .05$$

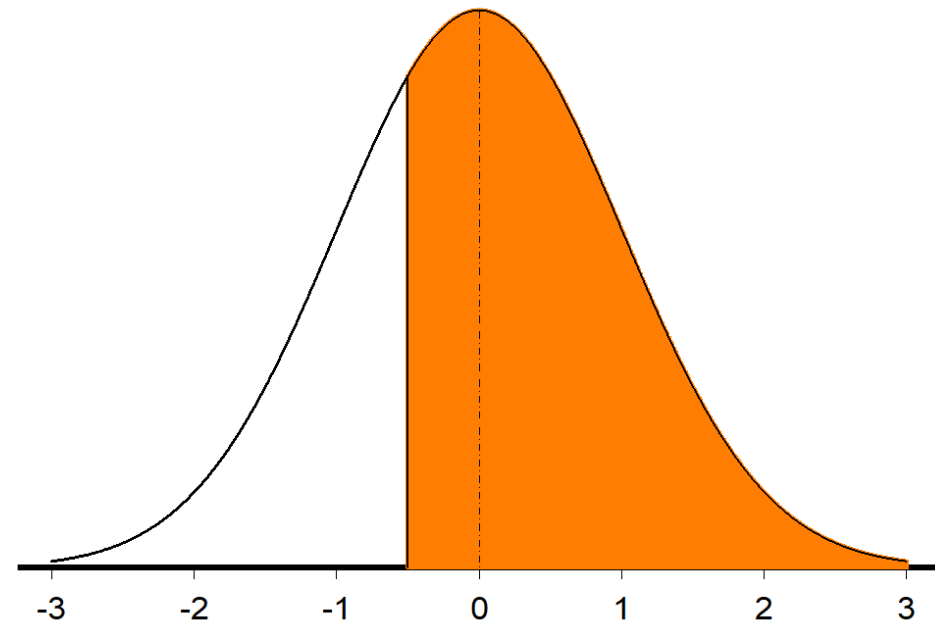
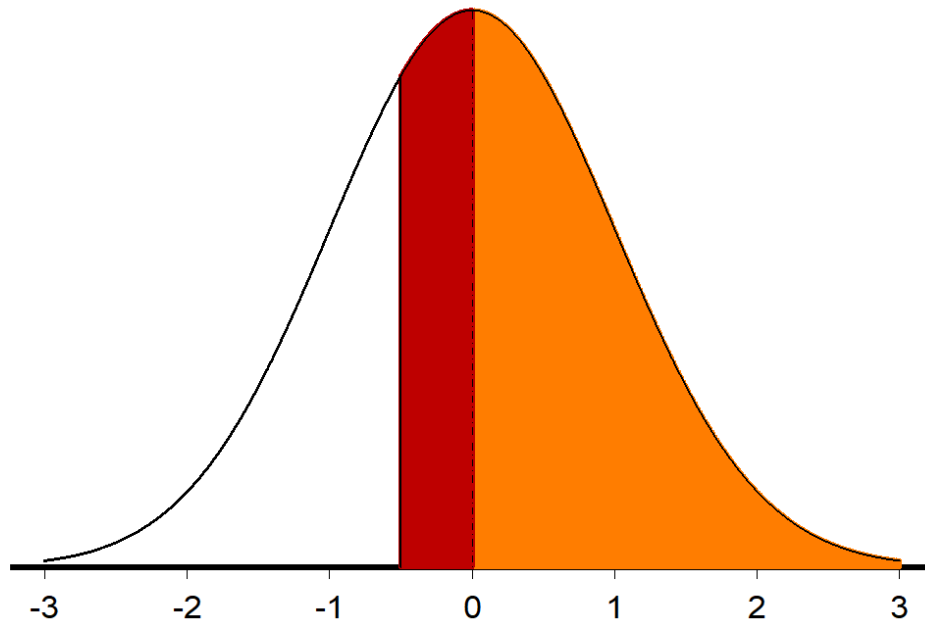


Properties of the Normal Distribution (3 / 4)

- Areas can be added

$$P(Z > 0) = .5$$

$$P(-.5 < Z < 0) = .19$$

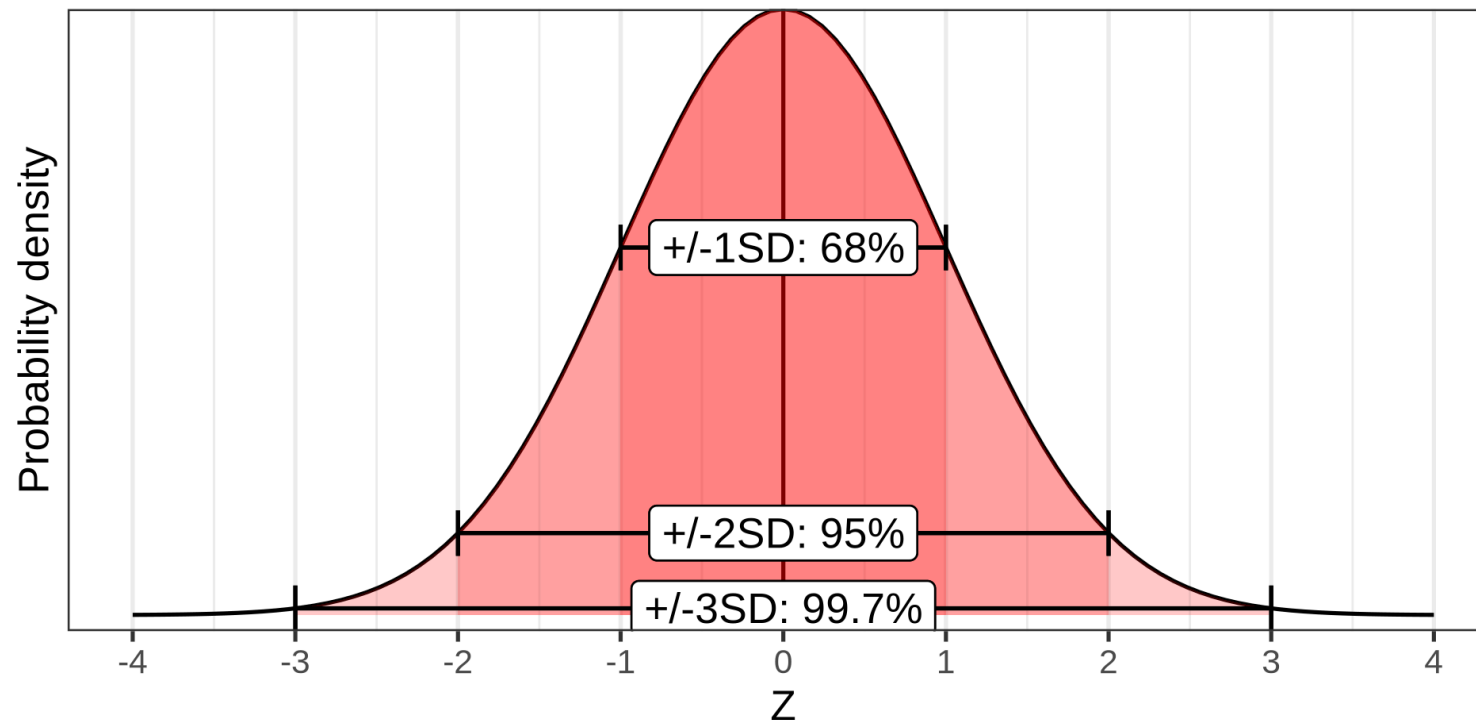




Properties of the Normal Distribution (4 / 4)

Standard percentages for mean, $\pm 1, 2, 3$ SD

- 50% of distribution is below μ , 50% above





Percentiles

The k -th percentile is the score below which k percent of scores fall

So in the standard normal distribution:

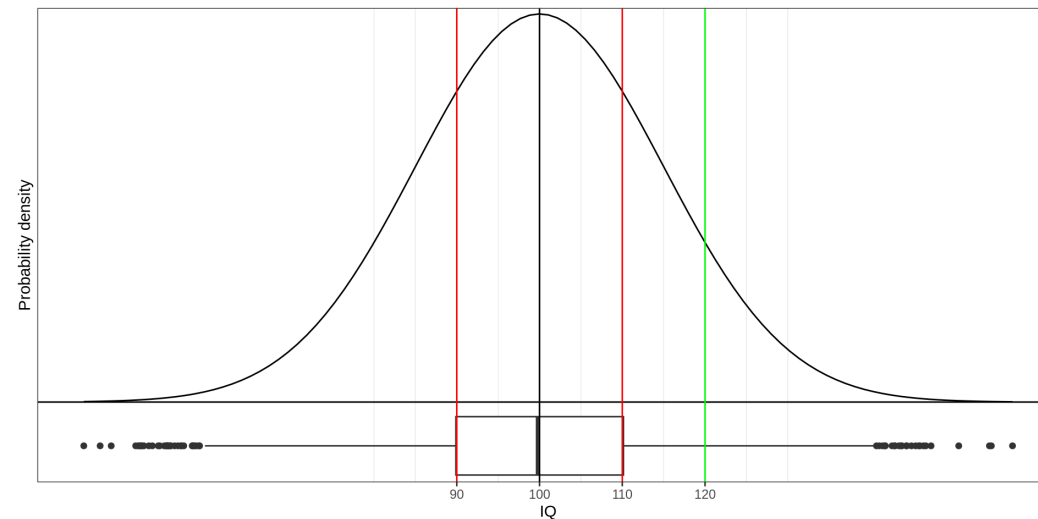
- The mean/median is the 50th percentile, because $P(Z < 0) = .5$
- We call the 25th and 75th percentile the first and third *quartile*
- +1SD is the 84th percentile, because $P(Z < 1) = .84$



Percentiles for X-scores

Let's apply this to $IQ \sim N(\mu = 100, \sigma = 15)$

- What percentile corresponds to $IQ < 120$?
- $Z = \frac{120-100}{15} = 1.33$
- $P(Z < 1.33) \approx .90$, so 90th percentile



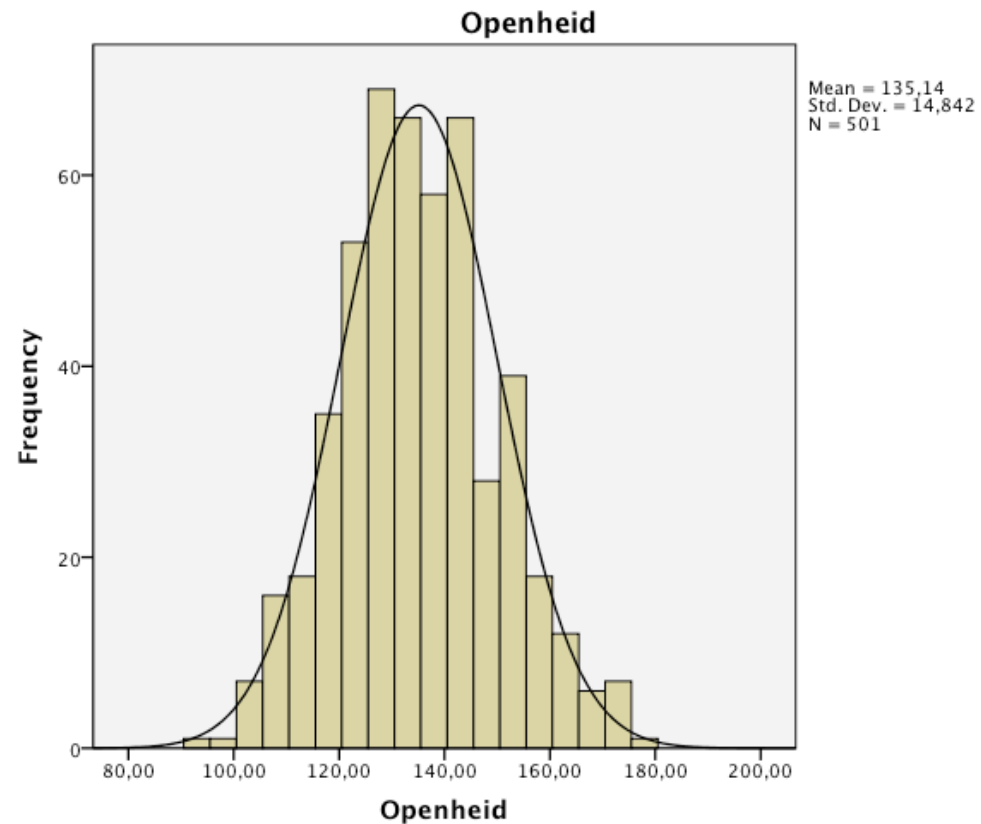


Describing data



A simple model

We can use the normal distribution as a *model* for the distribution of an observed variable





A simple model

“We assume variable X to be normally distributed”

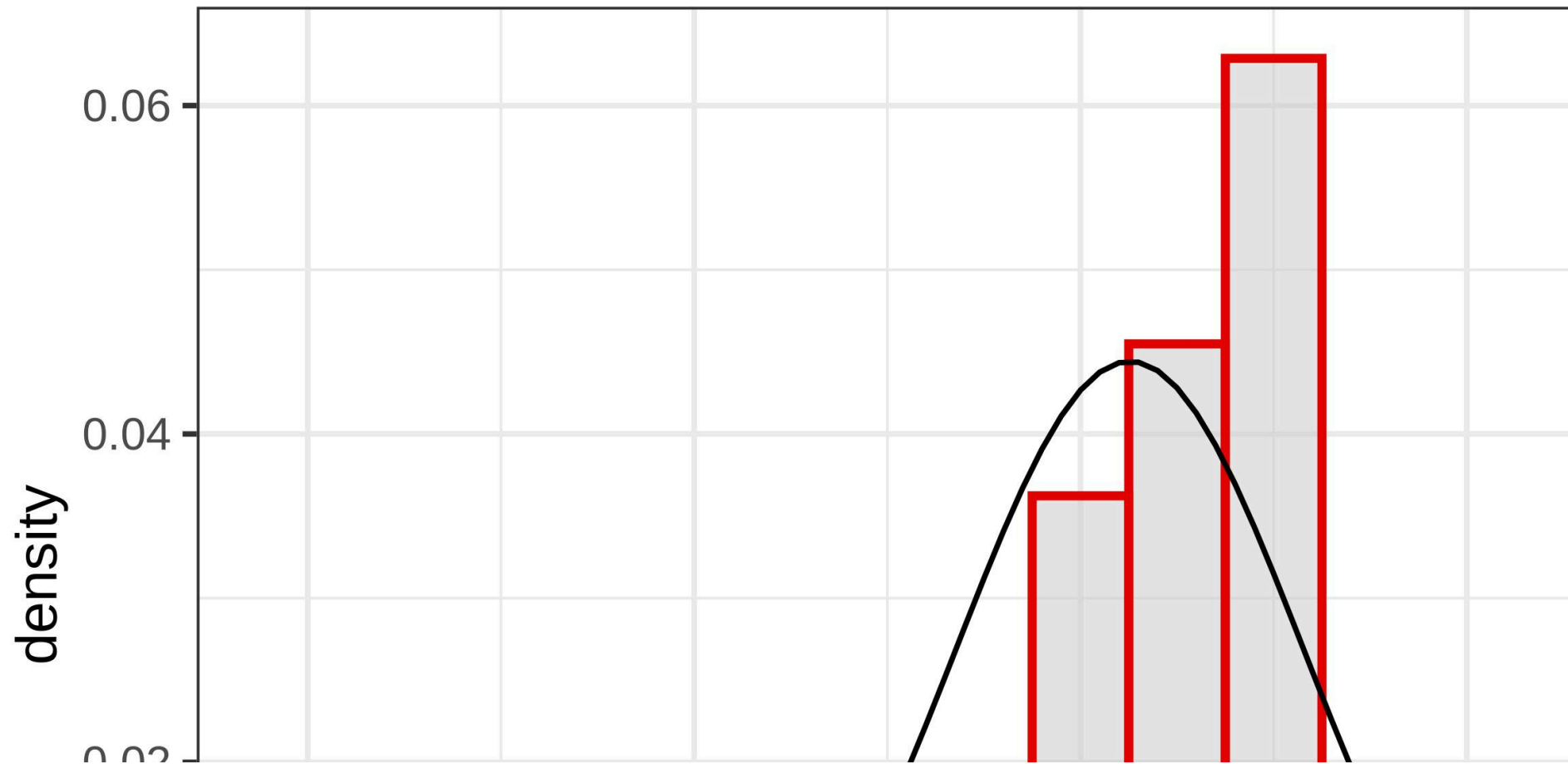
This allows us to:

- Summarize people's values with just a mean and SD
- Calculate probabilities of observing certain scores using table



Models can be wrong

Of course, the assumption can be wrong





What if the model is wrong?

Solution:

- Choose a different probability distribution (not part of this course)
- “Transform” the data to become “more normal”
- Comment that the assumption of normality may be violated (e.g., in Discussion section)